Comp 442 Group Assignment 2

“Phish Detector”

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Project & Dataset Overview

Project goal: develop a machine learning-based phishing and spam email detection system capable of classifying messages as either Spam or Ham.

Dataset Attributes

- Category - Indicates whether a message is spam or ham.
- Message - Contains the text content of the email.

Purpose of Analysis

- Understanding the dataset structure and quality
- Identify statistical properties of the data.
- Detect class imbalance, outliers, and distribution patterns.
- Prepare the dataset for future machine learning and NLP tasks.

Attribute	Value
Dataset Name	Email Spam Detection Dataset
Records	Kaggle
Records	5,572
Attributes	2
Classes	Spam and Ham
Format	CSV

Dataset Structure & Attributes

Category

- Indicates whether a message is **Spam** or **Ham (Legitimate)**.
- Serves as the **target variable** for supervised machine learning.
- Feature Type:
  - Categorical
  - Nominal
  - Binary

Dataset Format

- Stored in a structured CSV file.
- Organized into rows and columns.

Message

- Contains the text content of each email message.
- Serves as the primary **input feature** for classification.
- Feature Type:
  - Text Feature
  - Unstructured Data

Data Characteristics

- Structured at the dataset level.
- Contains unstructured text within the Message field.
- Requires Natural Language Processing (NLP) techniques before machine learning can be applied.

Data Quality Analysis

Missing values:

- Category: 0
- Message: 0

Finding:

- No missing values were detected in the dataset.
- No imputation or missing value treatment is required.

Duplicate records: 415

Finding**:**

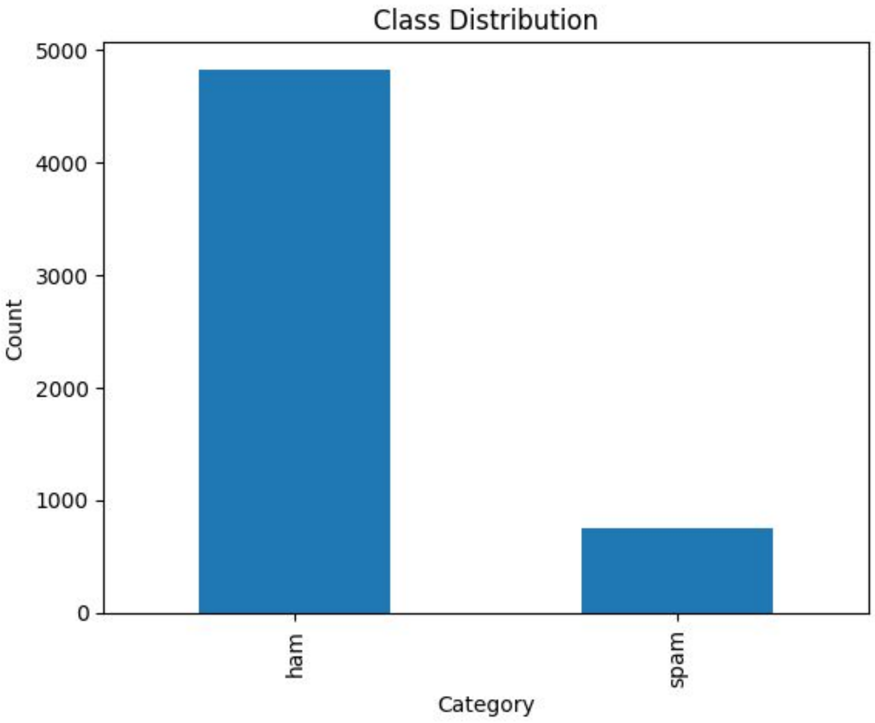
- 415 duplicate messages were identified.
- Duplicate records may introduce bias by overrepresenting certain message patterns.
- Duplicate messages should be reviewed and removed during preprocessing.

Class Distribution

Category	Count	Percentage
Ham	4,825	86.6%
Spam	747	13.4%

Key Observations

- Ham messages make up the majority of the dataset.
- Spam messages represent a much smaller portion of the data.
- The dataset exhibits class imbalance.



Message Length Statistics

Quartile	Value
Q1	35.75
Median	61
Q3	122
IQR	86.25

Statistic	Value
Mean	80.37
Median	61
Mode	22
Minimum	2
Maximum	910
Range	908
Standard Deviation	59.93

Key Observations

- The average message length is approximately 80 characters.
- Most messages contain fewer than 122 characters.
- Message lengths vary considerably across the dataset.
- The large range suggests the presence of unusually long messages.

Message Length Distribution

Histogram Analysis

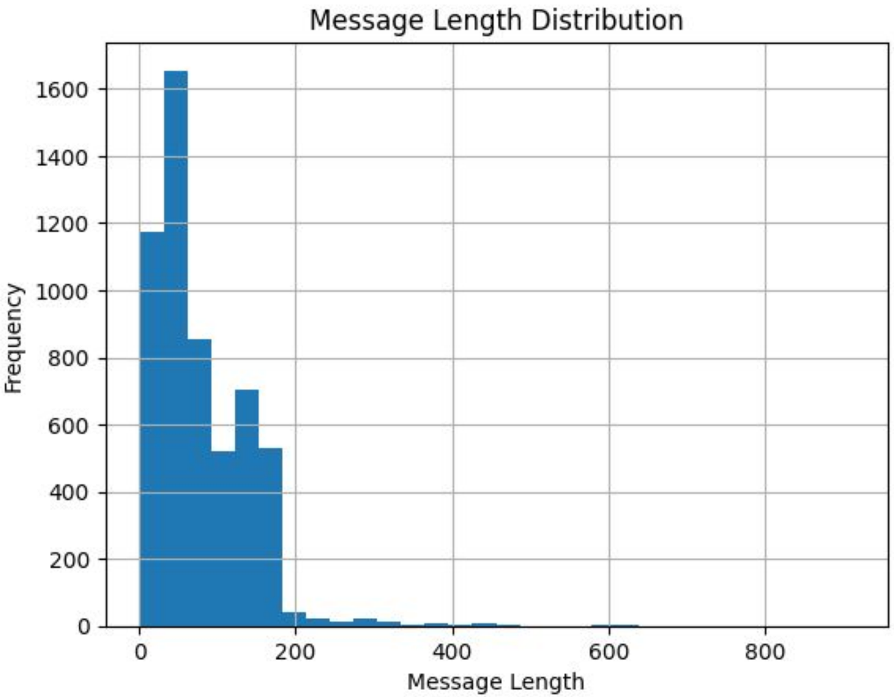
- Most messages are concentrated at shorter lengths.
- The distribution contains a small number of unusually long messages.
- The histogram shows a long tail extending to the right.

Key Findings

- The majority of messages contain fewer than 200 characters.
- Extremely long messages occur infrequently.
- The distribution is not symmetrical.

Interpretation

- Message length appears to be positively skewed.
- A small number of very long messages increase the overall average.
- These patterns suggest the presence of outliers that should be considered during preprocessing and model development.



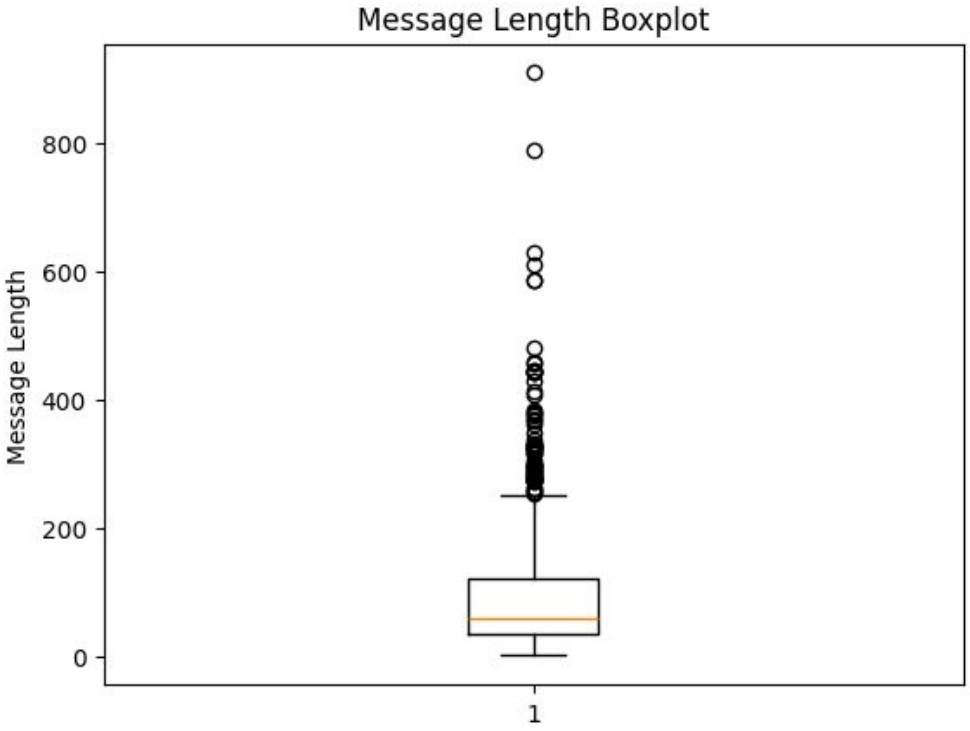
Message Length Boxplot

Boxplot Analysis

- The boxplot shows the spread of message lengths across the dataset.
- Most messages fall within the interquartile range (IQR) between 35.75 and 122 characters.
- Several data points appear above the upper whisker, indicating potential outliers.

Key Findings

- Multiple outliers are present in the dataset.
- Most messages are concentrated within a relatively small range.
- A small number of unusually long messages contribute to the overall spread of the data.
- The boxplot supports the positive skewness observed in the histogram.



Five-Number Summary

Minimum: 2 Q1: 35.75 Median: 61 Q3: 122 Max: 910

Word Count Statistics

Statistic	Value
Mean	15.58
Median	12
Mode	6
Minimum	1
Maximum	171
Range	170
Standard Deviation	11.41

Additional Statistics

- Variance: 130.11
- Skewness: 2.69

Quartile	Value
Q1	7
Median	12
Q3	23
IQR	16

Key Observations

- Most messages contain a relatively small number of words.
- The average message contains approximately 16 words.
- Word counts vary considerably across the dataset.
- The high skewness value indicates a strong positive skew caused by a small number of unusually long messages.

Correlation Analysis

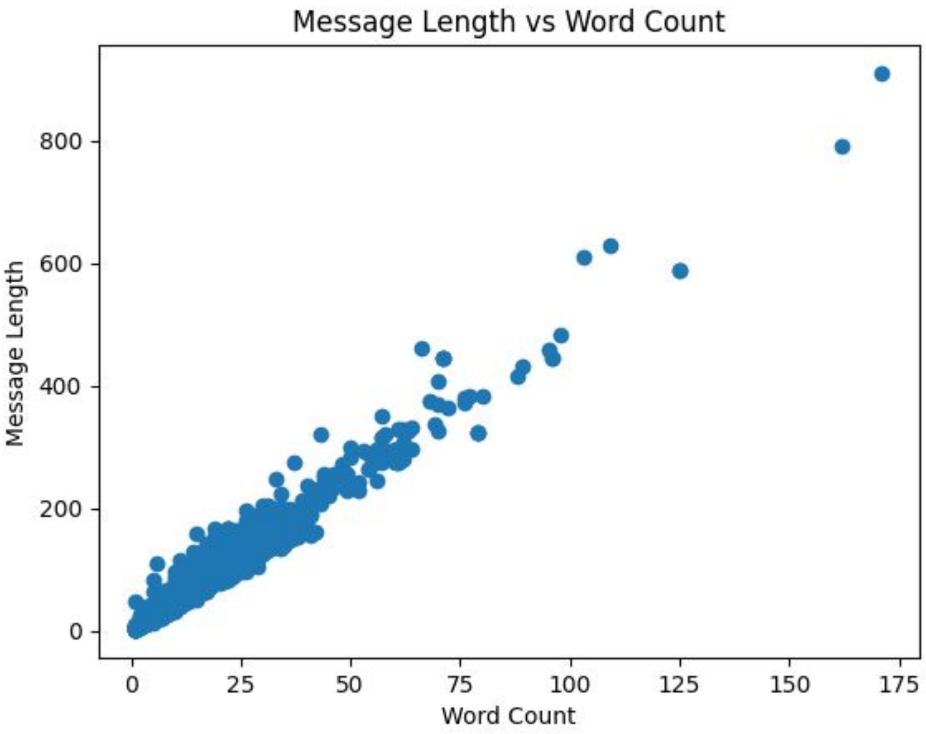
Correlation coefficient: 0.974

Interpretation

- The correlation coefficient of 0.974 indicates a very strong positive relationship between message length and word count.
- As the number of words in a message increases, the overall message length also increases.
- The relationship is nearly linear.

Scatter Plot Analysis

- The scatter plot shows a clear upward trend.
- Messages with higher word counts generally have greater message lengths.
- Few observations deviate significantly from the overall pattern.
- The plot visually confirms the strong correlation identified through statistical analysis.



Key Findings

- The dataset contains 5,572 labeled messages with no missing values.
- 415 duplicate records were identified and should be addressed during preprocessing.
- The dataset exhibits class imbalance, with ham messages representing 86.6% of the data.
- Both Message Length and Word Count show strong positive skewness.
- A very strong positive correlation (0.974) exists between Message Length and Word Count.

Challenges Identified

- Class imbalance may cause models to favor the majority class.
- Outliers are present in both message length and word count distributions.
- Text data requires preprocessing before machine learning can be applied.
- Duplicate messages may introduce bias if not removed.
- Some messages contain region-specific language and terminology, which may limit model generalizability.

Implications for Machine Learning

- Additional preprocessing and feature engineering will be necessary.
- Evaluation metrics such as Precision, Recall, and F1-Score should be used alongside accuracy.
- The extracted features provide meaningful information for spam classification.
- The dataset is suitable for supervised machine learning and NLP applications.

Conclusion
Summary

- The Email Spam Detection Dataset was successfully explored using statistical analysis techniques.
- The dataset contains 5,572 messages, no missing values, and a manageable number of duplicate records.
- Analysis identified class imbalance, positive skewness, and outliers in key features.
- A strong positive correlation (0.974) was observed between Message Length and Word Count.

Final Takeaways

- The dataset is suitable for supervised machine learning and spam classification tasks.
- Statistical exploration provided valuable insight into the dataset's structure and characteristics.
- The results of this analysis will support future preprocessing, feature engineering, model training, and evaluation for the Phish Detector project.